MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 15 (29 Nov-3 Dec 2026) - Period 1	Ch 8 - 8.12 Subroutines: scope and parameters

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 15 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Back from autumn break: retrieval quiz on loops and subroutines first. Before the break 75% wrote working functions; return vs print shaky.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use parameters (up to three) and distinguish local and global variables, explaining scope. Syllabus 8.1.6(b)-(c); AO1, AO2.

Success criteria — *what will successful learning look / sound like?*

I can pass up to three parameters to a subroutine; I can predict which variables are local or global; I can explain why local variables are safer.

Key knowledge / skill / understanding:

Parameters pass values in; local variables exist only inside the subroutine; global variables are accessible everywhere.

Likely misconception / access or challenge need:

Changing a local variable changes the global one with the same name; more parameters is always better.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*	ACTIVITY 2 *student task + evidence of learning*	ACTIVITY 3 *student task + evidence of learning*
Strategy: Gamification **Time:** 8 min	**Strategy:** Inquiry / investigation **Time:** 20 min	**Strategy:** Debate **Time:** 9 min
What students will do: Retrieval quiz after the break (Quizizz): 10 questions on loops, selection and subroutines.	**What students will do:** Scope investigation: run 4 Python programs with same-named local and global variables; predict, run, explain. Then write a function Discount(Price, Percent, IsMember) with 3 parameters.	**What students will do:** 'Global variables should be banned.' Teams argue using maintainability and testing evidence; class vote.
Evidence tool: Digital quiz (Forms / Kahoot / Quizizz) Assessment: ☑ Diagnostic ☐ Formative ☐ End Grouping: ☑ Individual ☐ Pair ☐ Group	**Evidence tool:** Portfolio / notebook work Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☐ Individual ☑ Pair ☐ Group	**Evidence tool:** Oral questioning Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☐ Individual ☐ Pair ☑ Group
Resources: Quizizz.	**Resources:** Coursebook (CUP 9781108915144) 8.12; scope programs; Python.	**Resources:** Debate prompt.
Support (low performers / SEN): Read-aloud mode.	**Support (low performers / SEN):** Scope diagram (boxes inside boxes).	**Support (low performers / SEN):** Sentence starters.
Stretch (fast learners / gifted): Explain the reason for 2 answers.	**Stretch (fast learners / gifted):** Use the global keyword and explain why it is discouraged.	**Stretch (fast learners / gifted):** Give a case where a global constant is acceptable.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Re-entry after the break: routines, success criteria. Model parameters with a function call and a scope diagram (global box containing subroutine boxes). Show local variable 'disappearing' after return.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students investigate scope experimentally and then debate a real software-engineering practice.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Expecting local changes to affect globals; wrong order of arguments; too many parameters; missing return. Likely: 'my global didn't change'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: identify local vs global in a 6-line program; write a 3-parameter procedure header. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Coursebook 8.12 scope questions; program: function that returns the average of three parameters.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Retrieval quiz showed good retention after the break (avg 72%). Scope experiments clarified a hard concept; debate was lively.

What will I change next time?

Allow more time for re-settling after the break; shorten debate.

Students needing follow-up:

Students below 50% on retrieval quiz (approx. 3 per class): targeted revision session.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☑C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 15 (29 Nov-3 Dec 2026) - Period 2	Ch 8 - 8.13 Library routines + 8.14 Maintainability

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 15 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Scope exit ticket: 75% identified local vs global. Final lesson of Term 1: library routines and making programs maintainable, then a mini-review.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use library routines MOD, DIV, ROUND, RANDOM and make programs maintainable with meaningful identifiers, comments and subroutines. Syllabus 8.1.7-8.1.8; AO2, AO3.

Success criteria — *what will successful learning look / sound like?*

I can use ROUND and RANDOM correctly in pseudocode and Python; I can improve an unreadable program with identifiers, comments and a subroutine; I can explain each improvement.

Key knowledge / skill / understanding:

Library routines are pre-written (ROUND, RANDOM); maintainable code is easy for others to read and update.

Likely misconception / access or challenge need:

RANDOM() returns whole numbers; comments make programs run slower; short identifiers are better.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 10 min

What students will do:

Dice simulator: Value <- ROUND(RANDOM() * 5, 0) + 1 (syllabus style), then Python random/round; discuss RANDOM range.

Evidence tool: Mini-whiteboards

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Syllabus p.42; Python random module.

Support (low performers / SEN):

Formula card for random ranges.

Stretch (fast learners / gifted):

Simulate 100 rolls and count each face (links to counting).

ACTIVITY 2 *student task + evidence of learning*

Strategy: Problem solving **Time:** 20 min

What students will do:

Maintainability makeover: a working but unreadable program (single-letter names, no comments, repeated code). Students refactor with meaningful identifiers, comments and one subroutine; mark against a checklist.

Evidence tool: Rubric

Assessment: ☐ Diagnostic ☐ Formative ☑ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Coursebook (CUP 9781108915144) 8.13-8.14; messy program file; rubric.

Support (low performers / SEN):

Checklist of 4 must-do improvements.

Stretch (fast learners / gifted):

Add a function and a constant; justify each change.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Cooperative learning **Time:** 7 min

What students will do:

Groups build a revision mind map of all Chapter 8 topics taught this term and share it on Teams.

Evidence tool: Self-assessment checklist

Assessment: ☐ Diagnostic ☐ Formative ☑ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Mind map template; Teams.

Support (low performers / SEN):

Template with branch names.

Stretch (fast learners / gifted):

Link each branch to a past-paper question type.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: two versions of the same program - which would you rather fix at 2 a.m.? Model library routines (DIV, MOD, ROUND, RANDOM) and the syllabus guidance on maintainable programs. Share rubric for the makeover.

What is new or innovative in this lesson? *(ETEC looks for this)*

A 'maintainability makeover' turns syllabus guidance into a practical refactoring task; groups produce a shared term revision map.

Progress recorded / shared:

☑ Gradebook ☑ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

RANDOM range errors; ROUND with no places; refactoring that breaks the program (test after each change). Likely: 'comments are only for the teacher'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Self-assessment (Forms): confidence for each Chapter 8 topic + one target for Term 2 (arrays and file handling next). Learners' questions.

Homework / platform *(e.g. Madrasati)*

Revision: past-paper Paper 2 questions on Topic 8.1 (selected); upload refactored program with comments.

5 REFLECT — After the lesson *Complete within a day. This is reflective-practice evidence and feeds your next Starting point.*

What worked — and how do I know?

Makeover task showed real understanding of maintainability; groups' mind maps were strong revision tools.
Self-assessments matched teacher judgements.

What will I change next time?

Give the makeover file earlier; some groups ran short of time for testing.

Students needing follow-up:

Students with low self-assessment confidence (approx. 3 per class): revision clinic before Semester 1 exams.

COVERAGE CHECK *Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.*

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☑C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H

Create	Discussion	Pair / group	Learner choice	Self-assessment	Peer assessment	SEN / low	Gifted
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☑C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback